

Reality Channels and Relational Propagation: A Foundational Framework for Emergent Reality and Geometry

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May 11, 2026

Abstract

This paper proposes a relational framework in which physical reality is interpreted as an emergent structure generated through interactions between systems across compatible *reality channels*. Instead of treating wavefunction collapse as an absolute global phenomenon, observation is reformulated as a process of local channel agreement between interacting systems.

Under this framework, shared classical reality emerges from the propagation and synchronization of compatible interaction channels across relational networks. We further propose that spacetime geometry itself may be understood as an emergent structure arising from paths and subgraphs within a graph of reality-channel interactions.

Deterministic geometry is interpreted as the effective geometry generated by dominant stable paths, while non-deterministic geometry arises from probabilistic subgraph structures containing multiple compatible paths. The framework is intended as a foundational ontological model rather than a complete predictive physical theory. Nevertheless, it provides a coherent conceptual structure connecting observation, entanglement, relational interaction, and emergent geometry.

Contents

1	Introduction	3
2	Conceptual Motivation	3
2.1	The Measurement Problem	3
2.2	Observation as Relational Agreement	3
2.3	Emergent Shared Reality	4
3	Refined Ontological Definitions	4
3.1	System	4
3.2	Reality Channel	4
3.3	Relational Compatibility	5
3.4	Interaction	5
3.5	Observation	5
3.6	Shared Reality	6
3.7	Reality Propagation	6

4	Refined Fundamental Postulates	6
4.1	Postulate I: Multi-Reality Postulate	6
4.2	Postulate II: Relational Channel Selection	7
4.3	Postulate III: Relational Propagation	7
4.4	Postulate IV: Emergent Geometry	7
5	Preliminary Relational Structure	7
5.1	Local Relational Consistency	7
5.2	Relativity of Relational Structure	8
5.3	Relational Domains	8
5.4	Channel Stabilization	8
5.5	Local Collapse Interpretation	8
6	Preliminary Theoretical Consequences	9
6.1	Theorem of Localized Reality	9
6.2	Theorem of Relational Non-Absoluteness	9
6.3	Theorem of Shared Reality Emergence	10
6.4	Theorem of Emergent Classicality	10
6.5	Theorem of Emergent Geometry	10
6.6	Theorem of Deterministic Geometry	10
6.7	Theorem of Non-Deterministic Geometry	11
6.8	Theorem of Relational Observation	11
6.9	Theorem of Ontological Relationality	11
7	Entanglement and Relational Structure	12
7.1	Relational Entanglement	12
7.2	Non-Local Relational Compatibility	12
7.3	Entanglement as Channel Agreement	12
7.4	Multipartite Relational Entanglement	13
7.5	Multiple Relational Channel Structures	13
7.6	Theorem of Multipartite Relational Branching	13
8	Reality-Channel Graphs	14
8.1	Reality-Channel Graph	14
8.2	Reality Paths	14
9	Emergent Geometry	14
9.1	Riemannian Geometry as Emergent Structure	14
9.2	Deterministic Geometry	14
9.3	Non-Deterministic Geometry	15
10	Open Problems	15
11	Conclusion	15

1 Introduction

The interpretation of quantum mechanics remains one of the deepest unresolved conceptual problems in modern physics. Standard quantum theory successfully predicts experimental outcomes with extraordinary precision, yet the ontological meaning of quantum states, measurement, and reality itself remains controversial.

In conventional formulations, a quantum system evolves continuously according to the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle, \quad (1)$$

while measurement appears to produce discontinuous state reduction into definite outcomes. This tension is commonly referred to as the measurement problem.

Several major interpretations have attempted to address this issue. The Copenhagen interpretation introduces wavefunction collapse as a special process associated with measurement. Everett's many-worlds interpretation removes collapse entirely and replaces it with branching universal states [1]. Relational Quantum Mechanics proposes that physical states are observer-relative rather than absolute [2]. Decoherence theory explains the suppression of interference through environmental interaction [3], although it does not completely resolve the ontological status of observed outcomes.

This paper proposes an alternative relational framework centered on the concept of *reality channels*. The central idea is that physical systems may exist within multiple possible channels of relational reality, while interactions between systems occur only through locally compatible channels. Observation is therefore interpreted not as an absolute collapse process, but as a synchronization or agreement of interaction channels between systems.

Within this framework, shared classical reality emerges from the large-scale propagation of compatible relational channels across networks of interacting systems. Furthermore, space-time geometry itself may be interpreted as an emergent structure generated from paths and connectivity relations within a graph of reality-channel interactions.

The purpose of this paper is not to present a finalized predictive theory, but rather to establish a foundational ontological and mathematical framework suitable for future refinement.

2 Conceptual Motivation

2.1 The Measurement Problem

Consider a quantum state represented by

$$|\psi\rangle = \sum_i c_i |i\rangle, \quad (2)$$

where the coefficients c_i represent probability amplitudes associated with possible outcomes.

Standard quantum mechanics predicts that measurement produces a single definite outcome. However, the mechanism by which a superposed state becomes a definite observed state remains conceptually unclear.

The present framework begins from the assumption that observed definiteness may emerge not from universal collapse, but from relational compatibility between interacting systems.

2.2 Observation as Relational Agreement

In the proposed framework, observation is fundamentally relational rather than absolute.

Instead of viewing measurement as a unilateral operation performed by an observer upon a system, we interpret observation as a process in which two interacting systems establish compatibility through a shared reality channel.

Observed definiteness therefore corresponds to local relational synchronization rather than destruction of alternative possibilities.

2.3 Emergent Shared Reality

If compatible channels propagate across sufficiently large interaction networks, stable domains of shared reality emerge. Classical reality may therefore be interpreted as a large-scale synchronization phenomenon.

3 Refined Ontological Definitions

The present framework is fundamentally relational. Consequently, the primary objects of the theory are not interpreted as isolated substances existing within a pre-defined spacetime background, but as relational structures capable of participating in coherent interaction.

The following definitions establish the ontological basis of the framework.

3.1 System

Definition 1 (System).

A system S is a relational entity capable of:

1. possessing multiple compatible or incompatible interaction possibilities,
2. participating in interaction with other systems,
3. contributing to the propagation of relational consistency.

A system is therefore not defined independently of interaction, but through its capacity to participate in relational structures.

3.2 Reality Channel

Definition 2 (Reality Channel).

A reality channel R is a relational compatibility structure governing the consistency of interaction outcomes between systems.

Equivalently, a reality channel defines a coherent mode through which systems may mutually realize interaction.

A reality channel is *not* interpreted as:

- a separate universe,
- a spatial dimension,
- a hidden variable state,
- or an independent classical world.

Instead, a reality channel represents a condition of coherent relational realization.

Formally, for a system S ,

$$\mathcal{R}(S) = \{R_1, R_2, \dots, R_n\} \tag{3}$$

denotes the set of compatible reality channels available to the system.

3.3 Relational Compatibility

Definition 3 (Relational Compatibility).

Two systems S_A and S_B are said to be relationally compatible under a channel R_k if coherent interaction between them may occur consistently within that channel.

Formally,

$$\mathcal{C}(S_A, S_B; R_k) = 1 \quad (4)$$

denotes compatibility between systems S_A and S_B under channel R_k .

If

$$\mathcal{C}(S_A, S_B; R_k) = 0, \quad (5)$$

then coherent interaction under R_k is forbidden.

Thus, interaction is constrained by relational compatibility rather than absolute global state reduction.

3.4 Interaction

Definition 4 (Interaction).

An interaction is a relational synchronization event between systems through a compatible reality channel.

Formally,

$$I(S_A, S_B) \Rightarrow R_k \in \mathcal{R}(S_A) \cap \mathcal{R}(S_B) \quad (6)$$

where R_k satisfies the compatibility condition

$$\mathcal{C}(S_A, S_B; R_k) = 1. \quad (7)$$

An interaction therefore does not globally eliminate alternative channels, but establishes local relational consistency.

3.5 Observation

Definition 5 (Observation).

Observation is a special case of relational interaction in which systems establish stable channel agreement.

Observation is therefore interpreted not as an absolute collapse process, but as local synchronization of relational consistency.

Define the relational agreement operator

$$\Gamma(A, B) \quad (8)$$

such that

$$\Gamma(A, B) : \mathcal{R}(A) \times \mathcal{R}(B) \rightarrow \mathcal{R}_{AB}, \quad (9)$$

where $\mathcal{R}_{AB}$ denotes the relational channel established through interaction.

Under this framework, observed definiteness corresponds to local channel stabilization rather than destruction of all alternative relational structures.

3.6 Shared Reality

Definition 6 (Shared Reality).

A shared reality domain is a connected relational structure generated through propagation of compatible reality channels across interacting systems.

Let

$$G_R = (V, E_R) \quad (10)$$

be the reality-channel graph.

A shared reality domain is defined as a connected subgraph

$$D \subseteq G_R \quad (11)$$

whose edges correspond to mutually compatible channel synchronizations.

Classical reality is interpreted as the large-scale stable limit of such domains.

3.7 Reality Propagation

Definition 7 (Reality Propagation).

Reality propagation is the extension of relational channel synchronization across connected interaction structures.

If

$$S_A \leftrightarrow S_B \quad (12)$$

establishes synchronization under channel R_k , and S_B subsequently interacts compatibly with S_C , then relational consistency may propagate:

$$R_k : S_A \rightarrow S_B \rightarrow S_C. \quad (13)$$

Shared classical reality therefore emerges as a propagation phenomenon rather than as a globally pre-defined structure.

4 Refined Fundamental Postulates

4.1 Postulate I: Multi-Reality Postulate

Every physical system possesses multiple possible relational channels of interaction.

Formally,

$$|S\rangle = \sum_i c_i |R_i\rangle, \quad (14)$$

where:

- $|R_i\rangle$ denotes a possible relational channel,
- c_i represents the relational amplitude associated with that channel.

The superposed structure represents relational interaction possibilities rather than simultaneous classical worlds.

4.2 Postulate II: Relational Channel Selection

Interactions between systems may occur only through mutually compatible reality channels.

Formally,

$$I(S_A, S_B) \Rightarrow \exists R_k \in \mathcal{R}(S_A) \cap \mathcal{R}(S_B) \quad (15)$$

such that

$$\mathcal{C}(S_A, S_B; R_k) = 1. \quad (16)$$

This interaction establishes local relational consistency without requiring universal collapse.

4.3 Postulate III: Relational Propagation

Compatible relational channels may propagate through connected interaction structures.

Large-scale propagation of compatible channels generates stable domains of shared classical reality.

Formally, if

$$R_k : S_1 \rightarrow S_2 \rightarrow \cdots \rightarrow S_n, \quad (17)$$

then the systems participating in this propagation belong to a common relational domain under channel R_k .

4.4 Postulate IV: Emergent Geometry

Spacetime geometry is not fundamental, but emerges from relational propagation structures within the reality-channel graph.

Deterministic geometry corresponds to dominant stable paths of propagation, while non-deterministic geometry corresponds to probabilistic subgraph structures containing multiple compatible paths.

5 Preliminary Relational Structure

The present framework introduces a relational ontology in which interaction consistency, rather than absolute state reduction, forms the basis of observable reality.

This section develops several preliminary formal consequences of the postulates introduced previously.

5.1 Local Relational Consistency

Proposition 1 (Local Consistency).

Let S_A and S_B be interacting systems.

If there exists a compatible channel R_k such that

$$\mathcal{C}(S_A, S_B; R_k) = 1, \quad (18)$$

then interaction between S_A and S_B establishes a locally consistent relational structure.

Interpretation.

Observed definiteness emerges locally through compatible synchronization rather than through universal elimination of alternative channels.

5.2 Relativity of Relational Structure

Proposition 2 (Relativity of Reality Channels).

Reality channels are relational rather than globally absolute.

Suppose systems S_A , S_B , and S_C interact.

Then it is possible for

$$R_{AB} \neq R_{AC}, \quad (19)$$

while both interactions remain locally consistent.

Interpretation.

A system may participate in distinct compatible relational structures relative to different interacting systems.

This implies that reality structure is fundamentally relational rather than globally fixed.

5.3 Relational Domains

Definition 8 (Relational Domain).

A relational domain is a connected region of the reality-channel graph whose systems share propagated channel compatibility.

Formally, let

$$D = (V_D, E_D) \subseteq G_R. \quad (20)$$

Then D is a relational domain if every edge in E_D corresponds to compatible synchronization under a common propagated channel structure.

Interpretation.

Classical reality corresponds to large stable relational domains.

5.4 Channel Stabilization

Definition 9 (Channel Stabilization).

A reality channel is stabilized within a relational domain if repeated compatible interactions preserve channel consistency across the domain.

Formally, let

$$R_k : S_1 \rightarrow S_2 \rightarrow \cdots \rightarrow S_n. \quad (21)$$

The channel R_k is stabilized if the propagation remains compatibility-preserving under subsequent interactions.

Interpretation.

Macroscopic classical stability emerges from large-scale stabilization of propagated channels.

5.5 Local Collapse Interpretation

Corollary 1 (Local Collapse).

Within the present framework, collapse corresponds to local channel stabilization rather than universal state reduction.

Interpretation.

A locally stabilized interaction appears observationally classical to participating systems, while broader external relational structures may remain unresolved or superposed.

6 Preliminary Theoretical Consequences

The following propositions and theorems are intended as preliminary formal consequences of the relational channel framework.

These statements are presently conceptual and foundational rather than rigorously derived mathematical theorems. Nevertheless, they provide structural insight into the implications of the framework.

6.1 Theorem of Localized Reality

Theorem 1 (Localized Reality Theorem).

Let S_A and S_B be interacting systems.

If interaction occurs through a compatible reality channel R_k , then observational definiteness emerges locally for systems participating in the interaction.

Formally,

$$I(S_A, S_B) \Rightarrow R_k \in \mathcal{R}(S_A) \cap \mathcal{R}(S_B) \quad (22)$$

with

$$\mathcal{C}(S_A, S_B; R_k) = 1. \quad (23)$$

Then the systems S_A and S_B establish locally stabilized relational consistency under R_k .

Sketch of Proof.

By Postulate II, coherent interaction requires existence of a compatible channel.

By Definition 5, observation corresponds to channel agreement.

Therefore, once interaction occurs, the participating systems inherit local relational consistency under the selected channel.

No assumption of universal global collapse is required.

□

6.2 Theorem of Relational Non-Absoluteness

Theorem 2 (Relational Non-Absoluteness).

Reality channels are not globally absolute structures.

A system may participate in distinct compatible relational structures relative to different interacting systems.

Formally,

there may exist systems S_A, S_B, S_C such that

$$R_{AB} \neq R_{AC}, \quad (24)$$

while both channel structures remain locally consistent.

Sketch of Proof.

By Definition 3, compatibility is defined relationally between interacting systems.

Compatibility therefore need not imply existence of a single globally shared channel structure.

Hence local relational consistency does not necessarily imply global relational uniqueness.

□

6.3 Theorem of Shared Reality Emergence

Theorem 3 (Emergence of Shared Reality).

Large-scale classical reality emerges from stable propagation of compatible relational channels across connected interaction domains.

Formally,

let

$$D = (V_D, E_D) \subseteq G_R \quad (25)$$

be a connected relational domain.

If channel compatibility remains stable across propagation paths within D , then systems within D experience effective shared reality.

Sketch of Proof.

By Postulate III, compatible channels may propagate through connected structures.

Repeated compatibility-preserving interactions stabilize relational synchronization.

In the large-scale limit, the domain behaves effectively as a coherent classical relational structure. □

6.4 Theorem of Emergent Classicality

Theorem 4 (Emergent Classicality).

Classical deterministic behavior emerges from dominance and stabilization of compatible relational paths.

Sketch of Proof.

By Definition 11, deterministic geometry corresponds to dominant stabilized reality paths.

As propagation stabilizes over large relational domains, fluctuations between incompatible channels become effectively suppressed.

Consequently, observational structure approaches classical deterministic behavior. □

6.5 Theorem of Emergent Geometry

Theorem 5 (Emergent Geometry Theorem).

Effective spacetime geometry emerges from propagation structures within the reality-channel graph.

Sketch of Proof.

Reality propagation generates connected relational structures represented by paths and subgraphs in G_R .

Stable path propagation produces effective continuity relations.

In the continuum approximation limit, these stable relational structures may be represented geometrically as manifold-like effective spacetime structures.

Thus, geometry is interpreted as emergent from relational propagation rather than fundamental. □

6.6 Theorem of Deterministic Geometry

Theorem 6 (Deterministic Geometry Theorem).

If relational propagation becomes strongly localized around a dominant compatible path, the resulting effective geometry becomes deterministic.

Formally,

let

$$P^* \subseteq G_R \quad (26)$$

be a dominant stabilized propagation path.

If compatible propagation probabilities outside P^* become negligible, then effective geometry reduces to a deterministic geometric structure generated by P^* .

Sketch of Proof.

When propagation overwhelmingly stabilizes around a single compatible path, relational uncertainty becomes effectively suppressed.

The resulting geometry therefore behaves classically and deterministically.

□

6.7 Theorem of Non-Deterministic Geometry

Theorem 7 (Non-Deterministic Geometry Theorem).

If multiple compatible propagation paths coexist within a relational subgraph, the resulting effective geometry becomes non-deterministic.

Formally,

let

$$H \subseteq G_R \quad (27)$$

contain multiple compatible propagation paths

$$P_1, P_2, \dots, P_n. \quad (28)$$

Then the effective geometry generated by H contains probabilistic relational structure.

Sketch of Proof.

Different compatible propagation paths correspond to alternative relational realizations.

Since no unique dominant stabilization exists, effective geometry inherits probabilistic structure from the subgraph connectivity relations.

□

6.8 Theorem of Relational Observation

Theorem 8 (Relational Observation Theorem).

Observation does not fundamentally correspond to projection-induced collapse, but to relational channel stabilization between interacting systems.

Sketch of Proof.

By Definition 5, observation is defined as channel agreement.

Observed definiteness therefore arises from local stabilization of compatible relational interaction.

Collapse becomes an emergent effective phenomenon associated with stabilized relational domains rather than a fundamental universal process.

□

6.9 Theorem of Ontological Relationality

Theorem 9 (Ontological Relationality Theorem).

Relational propagation structures are ontologically prior to effective spacetime geometry.

Sketch of Proof.

By Postulate IV, spacetime geometry emerges from relational propagation structures within the reality-channel graph.

Therefore, geometric structure depends on relational connectivity rather than relational connectivity depending on pre-existing geometry.

This reverses the conventional ontological ordering between geometry and interaction. $\square$

7 Entanglement and Relational Structure

Within the present framework, entanglement is interpreted not as the existence of a globally shared absolute state, but as a relational compatibility structure distributed across multiple interacting systems.

Under this interpretation, non-local correlation does not violate the relational ontology of the framework. Instead, entangled systems remain constrained by compatibility relations within shared or partially shared reality channels.

7.1 Relational Entanglement

Definition 10 (Relational Entanglement).

Two systems S_A and S_B are relationally entangled if their compatible interaction channels cannot be completely factorized into independent relational structures.

Formally, let

$$\mathcal{R}_{AB} \neq \mathcal{R}(S_A) \otimes \mathcal{R}(S_B), \quad (29)$$

where:

- $\mathcal{R}(S_A)$ and $\mathcal{R}(S_B)$ denote the individual channel structures,
- $\mathcal{R}_{AB}$ denotes the correlated relational structure shared between the systems.

Under this interpretation, entanglement corresponds to non-factorizable relational compatibility.

7.2 Non-Local Relational Compatibility

Proposition 3 (Non-Local Compatibility).

If systems S_A and S_B are relationally entangled, then compatible interaction constraints may remain correlated independently of local spatial separation.

Interpretation.

Non-local correlation is interpreted not as instantaneous transmission of physical information through spacetime, but as preservation of compatibility structure within a shared relational channel.

Thus, entanglement remains consistent with the relational ontology of the framework.

7.3 Entanglement as Channel Agreement

Within the present framework, entanglement itself is interpreted as a compatibility-preserving synchronization phenomenon.

Even in non-local interaction, entangled systems remain constrained by compatible realization within relational channels.

Thus, non-local correlation does not imply unrestricted interaction across all possible relational structures.

Instead, interaction remains restricted to mutually compatible channels.

This implies that entanglement is itself channel-selective.

7.4 Multipartite Relational Entanglement

Definition 11 (Multipartite Relational Entanglement).

Let

$$\mathcal{S} = \{S_1, S_2, \dots, S_n\} \quad (30)$$

be a multipartite relational system.

Multipartite relational entanglement occurs when the compatibility structure of the composite system cannot be reduced to independent pairwise relational channels.

Formally,

$$\mathcal{R}_{\mathcal{S}} \neq \bigotimes_{i=1}^n \mathcal{R}(S_i). \quad (31)$$

The multipartite system therefore possesses collective relational compatibility structures.

7.5 Multiple Relational Channel Structures

Postulate V (Multipartite Channel Multiplicity).

A multipartite entangled system may possess multiple compatible relational channel structures simultaneously.

Formally, for a multipartite system $\mathcal{S}$,

$$\mathcal{R}_{\mathcal{S}} = \{R_{\mathcal{S}}^{(1)}, R_{\mathcal{S}}^{(2)}, \dots, R_{\mathcal{S}}^{(m)}\}. \quad (32)$$

Different subsets of systems within $\mathcal{S}$ may therefore participate in distinct compatible relational channel configurations.

For example,

$$R_{ABC} \neq (R_{AB}, R_{AC}), \quad (33)$$

where:

- R_{ABC} denotes a globally shared multipartite relational channel,
- (R_{AB}, R_{AC}) denotes partially overlapping relational compatibility structures.

This implies that multipartite entanglement may generate branching relational compatibility configurations within the reality-channel graph.

7.6 Theorem of Multipartite Relational Branching

Theorem 10 (Multipartite Relational Branching).

If a multipartite entangled system possesses multiple compatible relational channel structures, then distinct stabilized relational propagation domains may emerge from different compatible channel configurations.

Sketch of Proof.

By Postulate V, multipartite systems may contain multiple compatible relational channel structures.

Different compatible synchronization pathways therefore generate distinct propagation structures within the reality-channel graph.

As relational propagation stabilizes, these structures may evolve into distinct effective relational domains.

Thus, multipartite entanglement introduces branching compatibility structures without requiring universal global collapse.

□

8 Reality-Channel Graphs

8.1 Reality-Channel Graph

Define a graph

$$G_R = (V, E_R), \quad (34)$$

where:

- V denotes systems or interaction events,
- E_R denotes reality-channel connections.

Nodes represent interacting systems, while edges represent synchronized relational channels.

8.2 Reality Paths

A reality path is defined as a connected sequence

$$P = (v_1, v_2, \dots, v_n) \subseteq G_R. \quad (35)$$

Such paths represent coherent propagations of compatible interaction channels.

9 Emergent Geometry

9.1 Riemannian Geometry as Emergent Structure

Conventional Riemannian geometry models spacetime as a smooth manifold equipped with local tangent spaces and geometric connection structures.

The present framework proposes a different ontological interpretation.

Instead of treating manifold structure as fundamental, we interpret manifold connectivity as an emergent projection of relational connectivity within the reality-channel graph.

Under this interpretation:

- local manifold patches correspond to locally synchronized relational domains,
- transitions between manifold patches correspond to node connections within reality-channel paths,
- geometric continuity emerges from propagating edge consistency in the graph.

Thus, when geometric connection occurs between local tangent structures, the interaction is not merely a connection between local neighborhoods, but also a synchronization between compatible reality channels.

Riemannian-like geometry may therefore be interpreted as the dynamical formation of edges along paths within the reality-channel graph.

9.2 Deterministic Geometry

Deterministic geometry is interpreted as geometry generated from dominant stable paths within the reality-channel graph.

Under this interpretation, classical spacetime corresponds to stable large-scale propagation along dominant relational paths.

9.3 Non-Deterministic Geometry

More generally, geometry may emerge not from a single path, but from probabilistic subgraphs containing multiple compatible paths.

Let

$$H \subseteq G_R \tag{36}$$

be a subgraph of compatible reality-channel structures.

The geometry generated by such a subgraph contains probabilistic connectivity relations between multiple possible paths.

This suggests that quantum uncertainty may correspond not only to uncertainty in physical state, but also to uncertainty in emergent geometric structure itself.

10 Open Problems

Several important problems remain unresolved:

1. the precise algebraic structure of reality channels,
2. compatibility relations between channels,
3. graph dynamics,
4. emergence of metric structure,
5. compatibility with relativistic quantum field theory,
6. possible empirical implications.

Future work may explore graph-theoretic, tensor-network, category-theoretic, and algebraic formulations of the framework.

11 Conclusion

This paper proposed a relational framework in which physical reality emerges through synchronization of compatible reality channels between interacting systems.

Observation was reformulated as channel agreement rather than absolute collapse, while shared classical reality was interpreted as large-scale propagation of compatible interaction channels.

The framework further proposed that spacetime geometry itself may emerge from path and subgraph structures within a graph of reality-channel interactions.

Although still preliminary, the framework provides a coherent ontological direction connecting quantum interaction, observation, entanglement, and emergent geometry within a unified relational interpretation.

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